Chapter 5

Standards, Constraints, and Milestones

5.1. Sustainability Standards

Sustainable development in the context of data-science and public health research requires that the outputs, methods, and data artefacts produced are designed to remain useful, reproducible, and ethically sound beyond the immediate scope of the project. This study aligns with the United Nations 2030 Agenda for Sustainable Development [39] across three goals that are directly relevant to suicide prevention and mental health research.

SDG 3 (Good Health and Well-being) — Target 3.4: The UN Sustainable Development Goal 3.4 calls on member states to reduce premature mortality from non-communicable diseases by one third and to promote mental health and well-being [40]. Suicide is explicitly tracked as an indicator under this target (SDG 3.4.2 — crude suicide rate per 100,000 population). The present study directly contributes to the evidence base required to pursue this target in Bangladesh by identifying high-risk demographic groups, geographic hotspots, and seasonally elevated risk periods — all of which are prerequisites for designing targeted prevention programmes.

SDG 5 (Gender Equality) — Target 5.2: The clustering analysis reveals that female individuals — particularly young married women and female students facing family conflict, academic pressure, and relationship-related distress — are consistently over-represented in the high-risk profile. SDG 5 calls for the elimination of all forms of violence against and exploitation of women and girls [39]. By providing empirically grounded, cluster-level evidence of gender-based suicide risk patterns, this study equips policymakers and NGOs with the evidence needed to design gender-sensitive mental health and suicide prevention interventions.

SDG 10 (Reduced Inequalities): The geographic and socioeconomic profiling conducted in this study reveals clear rural–urban disparities in suicide risk, consistent with the broader literature on inequality in Bangladesh [2, 18]. SDG 10 calls for reducing inequalities within and among countries [39]. The cluster-level identification of rural, low-income, and agriculturally dependent populations at elevated risk provides a data-driven basis for directing resources towards the most under-served communities.

Beyond alignment with the SDGs, the research was designed with long-term sustainability in mind in three additional ways. First, all code, preprocessing scripts, and the fully anonymised dataset will be made publicly available on GitHub under an open-source licence, enabling other researchers working in resource-constrained settings to replicate, extend, or adapt the methodology. Second, the modular 12-notebook pipeline architecture ensures that each component — data cleaning, imputation, clustering, evaluation — can be updated independently as new data become available or improved methods emerge. Third, by relying exclusively on free and open-source libraries (scikit-learn, pandas, NumPy, Geopy, Meteostat) and free computing resources (Google Colab), the framework is designed to be accessible to researchers in LMICs without institutional access to commercial software or high-performance computing infrastructure.

5.2. Impacts on Society

Real-world data-science research on sensitive public health topics such as suicide carries the potential for both significant positive social impact and, if not handled carefully, unintended negative consequences. This section examines both dimensions honestly.

Positive Impacts: The primary societal benefit of this research is its direct contribution to evidence-based suicide prevention in Bangladesh. Machine learning approaches applied to public health data have been shown to improve the precision of prevention strategies by identifying subgroups that would not emerge from aggregate-level analyses [41,42]. The cluster-specific risk profiles generated by this study — distinguishing, for example, between young female students in semi-urban areas experiencing family conflict and middle-aged rural males in financially distressed circumstances — enable different stakeholders to target their interventions with greater precision than has previously been possible with manually grouped Bangladeshi data. The following table summarises the expected impacts for specific stakeholder groups:

Stakeholder	Expected Impact
Ministry of Health and Family Welfare	Evidence-based identification of high-risk demographic groups and geographic districts to prioritise suicide prevention programmes and resource allocation.
NGOs and Mental Health Helplines	Cluster-specific risk profiles enable better targeting of outreach efforts, helpline staffing during high-risk seasons, and community awareness campaigns.
Schools and Universities	Insights into student-specific suicide triggers (academic pressure, exam periods, family conflict) can inform early identification and peer-support programme design.
Rural Development Agencies	Geographic risk profiling highlights high-burden rural districts, supporting the case for targeted livelihood and debt-relief interventions among at-risk populations.
Researchers and Data Scientists	The reproducible 32-pipeline framework and publicly released anonymised dataset provide a methodological foundation for future studies on incomplete social datasets in South Asia.
Media and Journalists	Empirical cluster findings provide a data-driven basis for responsible, non-sensational suicide reporting that highlights structural risk factors rather than individual narratives.

Table 5.1: Expected societal impacts by stakeholder group

Negative Impacts and Risks: The study also acknowledges potential negative societal consequences. Algorithmic bias is a well-documented risk in machine learning applications to public health data, particularly when training datasets are drawn from non-representative sources [43, 44]. Because newspaper-reported cases over-represent urban, female, and more unusual suicide incidents, any patterns identified through clustering reflect the biases of the source data rather than the true population distribution. Findings should therefore be interpreted as patterns within reported cases rather than as representative prevalence estimates.

A second risk concerns the potential misuse of cluster-level findings. Geographic risk profiling — while valuable for resource allocation — could, if communicated irresponsibly, contribute to stigmatisation of specific districts or demographic groups. This risk is mitigated in this study by framing all cluster descriptions in terms of structural risk factors (poverty, academic pressure, domestic conflict) rather than intrinsic characteristics of the populations concerned, in line with established ethical guidance on mental health data communication [45].

5.3. Ethical Considerations

Research involving suicide data is among the most ethically sensitive in the social sciences, requiring careful attention to privacy, non-maleficence, and the potential for harm to affected communities. This study adhered to the following ethical principles throughout:

Privacy and Anonymisation: The most immediate ethical obligation in working with individual-level suicide data is the prevention of re-identification. In this study, the full_name column was removed at the earliest stage of preprocessing, and the exact address field was standardised to district-level labels only — a deliberate design choice that reduces spatial precision while preserving analytical utility. The source URL (data_source column) was also permanently removed, preventing traceback to individual newspaper articles that might identify specific cases. These measures are consistent with best-practice guidance on the handling of psychiatric and behavioural research data [45].

Sensitive Topic Handling: Suicide data carries inherent risks of harm if specific case details — particularly method information — are reproduced or made easily accessible. In this study, method-of-suicide information is retained only in encoded form within the analytical dataset and is never reproduced at the individual case level in any output or visualisation. All cluster profiling is presented in aggregate form, consistent with responsible data communication practices for this domain [22].

Non-Stigmatisation: A persistent concern in suicide research is that algorithmic pattern-finding can inadvertently reinforce harmful stereotypes about particular demographic groups or communities. This study explicitly frames all cluster findings in terms of socioeconomic, environmental, and situational risk factors rather than personal or cultural failings. The language used in cluster labelling and profiling was reviewed to ensure it does not pathologise individuals or communities.

Data Source Legitimacy: All newspaper records used in this study were drawn from publicly available, freely accessible Bangladeshi news sources. No private medical records, police files, or non-public databases were accessed. The use of publicly available media content for academic suicide research is an established and ethically accepted practice in the field, as documented in the broader literature on media-based suicide data collection in LMICs [22].

Beneficence Principle: The overarching ethical justification for this research is its potential to contribute to life-saving prevention efforts. The research was conducted with the primary goal of generating actionable insights that can reduce suicide mortality in Bangladesh, not merely to advance academic knowledge. This beneficence orientation shaped all methodological decisions, including the choice to prioritise interpretable cluster solutions over those with marginally higher geometric quality metrics.

Algorithmic Fairness: Machine learning models trained on biased data can systematically disadvantage under-represented groups [43, 44]. To mitigate this risk, the balance metric introduced in this study explicitly penalises clustering solutions that produce highly unequal cluster sizes — a design choice that promotes fairer representation of minority subgroups in the final analysis. Additionally, all imputation and clustering results were examined across demographic subgroups (gender, age group, geographic region) to check for systematic distortions before cluster profiles were finalised.

5.4. Challenges Faced

The implementation of this research involved numerous technical, methodological, and logistical challenges. The following table summarises the most significant challenges encountered and the solutions applied:

Challenge Encountered	Solution Applied
Extreme missingness (>90%) in several key columns such as previous mental health history and family size	Columns exceeding 70% missingness permanently excluded; remaining columns treated with three advanced imputation methods under two predictor strategies
Inconsistent Bangla–English spellings and over-granular categorical labels across 13 columns	Iterative manual cleaning and semantic harmonisation; district-level standardisation of hometown field reducing cardinality from 594 to 151 unique values
No reliable geocoding service for rural Bangladeshi locations	Geopy geocoding API with a cached JSON fallback system; reduced latitude/longitude missingness from ~53% to 6.49%
Weather API rate limits and missing historical records for remote locations	Hourly-to-daily fallback strategy with extensive local caching; partial reduction in weather-variable missingness achieved
High computational cost of MissForest imputation (100 trees × 48 features × 1,617 rows, repeated per variant)	Cached intermediate outputs and modular notebook structure that allowed each variant to be executed and re-run incrementally on the free Google Colab tier without losing prior progress
Standard clustering validity metrics favoured geometrically tight but practically useless solutions (e.g. DBSCAN with 99% noise)	Introduction of a novel balance metric weighted at 35% in the composite score to penalise imbalanced and noise-heavy solutions
Choosing the right encoding strategy for 13 heterogeneous categorical variables with widely different cardinalities	Four encoding strategies applied contextually: ordinal for ordered features, one-hot for low-cardinality nominals, frequency encoding for high-cardinality nominals, and label encoding for MissForest compatibility

Table 5.2: Key challenges encountered and solutions applied during the research

Among these challenges, the most consequential was the evaluation problem: the observation that certain DBSCAN configurations produced silhouette scores above 0.87 while simultaneously clustering only 6 records out of 1,617 — a technically high but practically worthless result. This finding, encountered during early manual inspection of the 32 pipeline outputs, directly motivated the introduction of the balance metric and was the defining methodological insight that shaped the entire evaluation framework. It underscores a broader lesson about the dangers of relying on a single metric in unsupervised learning evaluation, particularly in applied social science contexts where interpretability is as important as geometric optimality.

5.5. Constraints

In addition to the technical challenges described above, the research operated under several structural constraints that defined the boundaries of what was achievable within the scope of this project:

Constraint Type	Description
Data	Limited to newspaper-reported completed suicides only. Suicide attempts, unreported cases, and cases from clinical or police records are not captured, introducing systematic under-coverage bias.
Geographical	Rural areas are under-represented because local and vernacular newspapers are harder to access digitally, and remote incidents receive less media coverage.
Temporal	Data spans 2017–2025; long-term secular trends and pre-2017 patterns cannot be analysed. The dataset is also not updated in real time.
Methodological	The study is limited to unsupervised pattern discovery. No supervised prediction model is developed, and no external ground-truth labels are available to validate cluster assignments.
Hardware and Budget	The entire pipeline was executed using free and student-tier resources (Google Colab, GitHub). MissForest training required 4–6 hours per run, limiting the number of hyperparameter combinations that could be explored.
Language	Free-text columns in Bangla (e.g. profession_description, reason_description) could not be processed with standard NLP tools and were excluded from the analysis.

Table 5.3: Structural constraints affecting the scope and generalisability of the research

The most consequential of these constraints is the data limitation: because the dataset is drawn exclusively from newspaper-reported completed suicides, the findings cannot be generalised to suicide attempts, unreported deaths, or cases that received no media coverage. This structural bias is an inherent feature of all newspaper-based suicide research in LMICs [22] and represents the primary caveat that any policymaker or practitioner using these findings should bear in mind.

5.6. Project Timeline and Milestones

The research was conducted over a total of 25 weeks. The following table presents the actual project timeline, organised by week, with the corresponding task and completion status for each milestone:

Week(s)	Task	Status
1–3	Literature review, identification of research gap, and dataset collection from Kaggle	Completed
4–7	Manual collection of 857 additional cases from verified Bangladeshi news sources (2017–2025); addition of 9 new socioeconomic and clinical attributes	Completed
8–10	Intensive data cleaning: standardisation of column names, date formats, categorical harmonisation, and removal of irrelevant columns	Completed
11–12	Conditional filling using Geopy geocoding API (latitude, longitude) and Meteostat meteorological API (weather variables)	Completed
13–14	Feature engineering, missing value analysis, and multi-strategy encoding (ordinal, one-hot, frequency, label)	Completed
15–18	Implementation of six imputation variants (MICE, KNN, MissForest $\times$ 2 predictor strategies) with post-imputation processing and validation	Completed
19–21	Clustering experiments: K-Means, Agglomerative, DBSCAN, Spectral Clustering across all 8 datasets (32 pipeline combinations); metric computation	Completed
22–23	Composite scoring, ranking, group-level comparisons, and detailed cluster profiling of the best-performing pipeline (agg_rf_no_context)	Completed
24–25	Report writing, figure production, reference compilation, and final documentation	Completed

Table 5.4: Project timeline and milestone completion status (Total duration: 25 weeks)

The most time-intensive phases were the manual data collection (weeks 4–7) and the imputation implementation (weeks 15–18). The former required careful verification of each new case against multiple source articles to ensure factual accuracy and avoid duplication. The latter required not only the implementation of three distinct imputation algorithms under two predictor strategies each, but also the development of custom post-imputation processing logic — including one-hot group correction and ordinal rounding — to ensure that all six imputed datasets were analytically coherent before being passed to the clustering stage.

The clustering experiments (weeks 19–21) were computationally the most demanding phase, requiring grid search and fine-tuning across up to 20 parameter combinations per dataset per algorithm, run independently across all 8 input datasets. The final evaluation and profiling phase (weeks 22–23) was the most analytically intensive, requiring the design and implementation of the composite scoring framework, group-level comparisons, and full decoding and interpretation of the best-performing cluster solution.